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Math 208 Final Exam Term 221

Original exam date: 24 December 2022. Practice version: questions and choices shuffled.
Net time allowed: 180 minutes

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1. A possible fundamental matrix for the system $X' = \begin{bmatrix} 4 & 2 \\ 3 & -1 \end{bmatrix} X$ is

(a) $\Phi(t) = \begin{bmatrix} 3e^{-2t} & 0 \\ 5e^{-2t} & e^{5t} \end{bmatrix}$

(b) $\Phi(t) = \begin{bmatrix} e^{-2t} & 2e^{5t} \\ -3e^{-2t} & e^{5t} \end{bmatrix}$

(c) $\Phi(t) = \begin{bmatrix} e^{-2t} & e^{5t} \\ e^{-2t} & 2e^{5t} \end{bmatrix}$

(d) $\Phi(t) = \begin{bmatrix} e^{-2t} & 2e^{5t} \\ 0 & e^{5t} \end{bmatrix}$

(e) $\Phi(t) = \begin{bmatrix} e^{-2t} & e^{5t} \\ -3e^{-2t} & 0 \end{bmatrix}$

2. If $X = c_1 \begin{bmatrix} 5 \\ -6 \end{bmatrix} e^{3t} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{4t}$ is the solution of the initial value problem

$$X' = \begin{bmatrix} 9 & 5 \\ -6 & -2 \end{bmatrix} X, \quad X(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

then $c_2^2 - c_1^2 =$

(a) 35

(b) 36

(c) 37

(d) 32

(e) 34

3. If $(x, y, z) = (a, b, c)$ is the solution of the system

$$\begin{aligned} 2x + 8y + 3z &= 2 \\ x + 3y + 2z &= 5 \\ 2x + 7y + 4z &= 8 \end{aligned}$$

then $a + b + c =$

(a) 5

(b) 6

(c) 0

(d) 7

(e) 3

4. Using variation of parameters to find a particular solution X_p of the nonhomogeneous system $X' = AX + \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^t$ where $X_c = c_1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^t + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{2t}$ form a general solution of the associated homogeneous system, then $X_p(1) =$

(a) $\begin{bmatrix} 0 \\ 3e \end{bmatrix}$

(b) $\begin{bmatrix} e \\ 5e \end{bmatrix}$

(c) $\begin{bmatrix} 7e \\ e \end{bmatrix}$

(d) $\begin{bmatrix} 3e \\ 2e \end{bmatrix}$

(e) $\begin{bmatrix} 7e \\ 5e \end{bmatrix}$

5. Let $A = \begin{bmatrix} 1 & -2 & 1 \\ 0 & 1 & 0 \\ 0 & -2 & 2 \end{bmatrix}$. Using Cayley-Hamilton Theorem, $A^4 = aA^3 + bA^2 + cA$. Then $a + b + c =$

(a) 2

(b) 4

(c) 1

(d) 0

(e) 3

6. Let

$$A = \begin{bmatrix} 0 & 3 & 4 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

If $e^{At} = \begin{bmatrix} 1 & f(t) & h(t) \\ 0 & 1 & 3t \\ 0 & 0 & 1 \end{bmatrix}$, then $f(2) + h(2) =$

(a) 24

(b) 32

(c) 30

(d) 28

(e) 26

7. By using the method of variation of parameters, a particular solution y_p of the differential equation

$$y'' - 9y = \frac{9x}{e^{3x}}$$

is

(a) $y_p = \frac{1}{4}x^3e^{-3x} + \frac{1}{4}xe^{-3x} - \frac{1}{24}e^{-3x}$

(b) $y_p = \frac{3}{4}x^2e^{-3x} + \frac{1}{2}xe^{-3x} - \frac{1}{24}e^{-3x}$

(c) $y_p = \frac{1}{2}x^2e^{-3x} + \frac{1}{4}x^3e^{-3x} - \frac{1}{24}e^{-3x}$

(d) $y_p = -\frac{3}{4}x^2e^{-3x} - \frac{1}{4}xe^{-3x} - \frac{1}{24}e^{-3x}$

(e) $y_p = -\frac{3}{4}xe^{-3x} + \frac{1}{4}xe^{-3x} - \frac{1}{24}e^{-3x}$

8. Let $A = \begin{bmatrix} 9 & -8 \\ 6 & -5 \end{bmatrix}$. A diagonalization matrix P, such that $P^{-1}AP$ is diagonal is

(a) $P = \begin{bmatrix} -1 & 4 \\ 1 & -3 \end{bmatrix}$

(b) $P = \begin{bmatrix} 1 & -2 \\ 2 & 5 \end{bmatrix}$

(c) $P = \begin{bmatrix} 1 & 2 \\ 1 & 5 \end{bmatrix}$

(d) $P = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$

(e) $P = \begin{bmatrix} 1 & 4 \\ 1 & 3 \end{bmatrix}$

9. The differential equation

$$t^3x''' - 2t^2x'' + 3tx' + 5x = \ln t$$

is equivalent to the system of first-order equations.

(a) $x_1' = x_2,\ x_2' = x_3,\ x_3' = -5x_1 - 3tx_2 - 2x_3 + \ln t$

(b) $x_1' = x_2,\ x_2' = x_3,\ t^3x_3' = -5x_1 - 3tx_2 + 2t^2x_3 + \ln t$

(c) $x_1' = x_2',\ x_2' = x_1,\ t^3x_3' = 5x_1 + 3tx_2 + 2t^2x_3 + \ln t$

(d) $x_1' = x_2,\ x_2' = x_3,\ x_3' = -5x_1 - 3tx_2 + 2t^2x_3 - \ln t$

(e) $x_1' = x_1,\ x_2' = x_2,\ x_3' = -5x_1 - 3tx_2 + 2t^2x_3 + \ln t$

10. The general solution of the first order homogeneous system $X' = \begin{bmatrix} 1 & 2 & 1 \\ 6 & -1 & 0 \\ -1 & -2 & -1 \end{bmatrix} X$ is given by

$$X = c_1 \begin{bmatrix} a \\ b \\ -2 \end{bmatrix} e^{3t} + c_2 \begin{bmatrix} \alpha \\ \beta \\ 1 \end{bmatrix} e^{\lambda t} + c_3 \begin{bmatrix} e \\ f \\ 13 \end{bmatrix} e^{\mu t}$$

then $a \cdot b \cdot \lambda =$

(a) -12

(b) -24

(c) 24

(d) 0

(e) 12

11. The matrix $A = \begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & 1 \\ 1 & -1 & -1 \end{bmatrix}$ has only one eigenvalue $\lambda = -1$ which is defective of defect 2. If we choose $v_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ such that $(A+I)^3 v_3 = 0$, and $(A+I)^2 v_3 \neq 0$, then the general solution of $X' = AX$ is

(a) $X = \left(c_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} t \\ t \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} 1 - \frac{t^2}{2} \\ \frac{t^2}{2} \\ t \end{bmatrix} \right) e^{-t}$

(b) $X = \left(c_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} t \\ t \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} \frac{t^2}{2} + 1 \\ \frac{t^2}{2} \\ t \end{bmatrix} \right) e^{-t}$

(c) $X = \left(c_1 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} t \\ -t \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} \frac{t^2}{2} + 1 \\ t^2 \\ t \end{bmatrix} \right) e^{-t}$

(d) $X = \left(c_1 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ t \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} \frac{t^2}{2} \\ \frac{t^2}{2} \\ t \end{bmatrix} \right) e^{-t}$

(e) $X = \left(c_1 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ t \\ t \end{bmatrix} + c_3 \begin{bmatrix} 1 + \frac{t^2}{2} \\ \frac{t^2}{2} \\ t \end{bmatrix} \right) e^{-t}$

12. The solution of $X' = \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix} X$, $X(0) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ at $t = \dfrac{\pi}{4}$ equals

(a) $\begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{-\pi/4}$

(b) $\begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{\pi/4}$

(c) $\begin{pmatrix} -2 \\ 1 \end{pmatrix} e^{\pi/4}$

(d) $\begin{pmatrix} -2 \\ 0 \end{pmatrix} e^{\pi/4}$

(e) $\begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{\pi/4}$

13. Let S be a subspace of $\mathbb{R}^4$ defined by $S = \{(a, b, c, d) \mid a = b + c + d\}$. A basis for the subspace is

(a) $\{(1, 1, 0, 0), (1, 0, 1, 0), (1, 0, 0, 1)\}$

(b) $\{(1, 1, 0, 0), (1, 0, 1, 0), (2, 0, 1, 2)\}$

(c) $\{(1, 1, 1, 0), (1, 0, 1, 0), (1, 0, 1, 1)\}$

(d) $\{(1, 1, 0, 0), (1, 0, 1, 0)\}$

(e) $\{(1, 0, 1, 0), (1, 0, 0, 1)\}$

14. The general solution of the differential equation

$$y''' + 3y'' - 4y = 0$$

is

(a) $y(x) = c_1e^{-x} + c_2e^{3x} + c_3xe^{3x}$

(b) $y(x) = c_1e^{x} + c_2e^{-2x} + c_3e^{3x}$

(c) $y(x) = c_1e^{x} + c_2e^{-2x} + c_3e^{4x}$

(d) $y(x) = c_1e^{x} + c_2e^{-2x} + c_3xe^{-2x}$

(e) $y(x) = c_1e^{-x} + c_2e^{-2x} + c_3xe^{-2x}$

15. If $y(x)$ is the solution of the initial value problem

$$\frac{dy}{dx} = \frac{2x + 1}{2y}, \qquad y(-2) = -1,$$

then $y(2) =$

(a) $-\sqrt{3}$

(b) 0

(c) $\sqrt{5}$

(d) $-\sqrt{5}$

(e) $\sqrt{3}$

16. The characteristic equation of a matrix A is $(\lambda+1)(\lambda-5)^3 = 0$, where we have only two linearly independent eigenvectors corresponding to $\lambda = 5$. The Jordan normal form of A is

(a) $\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 5 & 1 & 0 \\ 0 & 0 & 5 & 1 \\ 0 & 0 & 0 & 5 \end{bmatrix}$

(b) $\begin{bmatrix} 5 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$

(c) $\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}$

(d) $\begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 5 & 1 \\ 0 & 0 & 0 & 5 \end{bmatrix}$

(e) $\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 5 & 1 \\ 0 & 0 & 0 & 5 \end{bmatrix}$

17. The general solution of the differential equation

$$x\frac{dy}{dx} - 3y = x^3$$

is

(a) $y = x^3 \ln x + cx^2$

(b) $y = x^3 \ln x + cx$

(c) $y = x^3 \ln x + cx^3$

(d) $y = x^2 \ln x + cx^2$

(e) $y = x^2 \ln x + cx^3$

18. Let $A = \begin{bmatrix} 3 & 5 & -2 \\ 0 & 2 & 0 \\ 0 & 2 & 1 \end{bmatrix}$. An eigenvector corresponding to the eigenvalue $\lambda = 2$ of A is

(a) $\begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$

(b) $\begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$

(c) $\begin{bmatrix} -1 \\ 3 \\ 2 \end{bmatrix}$

(d) $\begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$

(e) $\begin{bmatrix} -1 \\ 1 \\ 3 \end{bmatrix}$

19. The general solution of the exact differential equation

$$(3x^2 + 2y^2)\,dx + (4xy + 6y^2)\,dy = 0$$

is

(a) $x^3 - 2xy^2 - 2y^3 = c$

(b) $x^3 - 2x^2y - 2y^3 = c$

(c) $x^3 + 2xy^2 + 2y^3 = c$

(d) $x^3 - 2xy^2 + 2y^3 = c$

(e) $x^3 - 2x^2y + 2y^3 = c$

20. If $y(x)$ is the solution of the initial-value problem

$$y'' + 4y = 2x; \quad y(0) = 1,\ y'(0) = 2$$

then $y(\pi) =$

(a) $1 - \dfrac{\pi}{2}$

(b) $1 + \dfrac{\pi}{4}$

(c) $1 + \dfrac{\pi}{3}$

(d) $1 + \dfrac{\pi}{2}$

(e) $1 - \dfrac{\pi}{4}$

Answer Key

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Q	Ans	Brief explanation
1	**(b)**	$\lambda = -2 : (1,-3)$, $\lambda = 5 : (2,1)$; $\Phi = [e^{-2t}(1,-3) \;\; e^{5t}(2,1)]$.
2	**(a)**	$5c_1 + c_2 = 1$, $6c_1 + c_2 = 0$: $c_1 = -1$, $c_2 = 6$, so $36 - 1 = 35$.
3	**(a)**	Eliminate x: $y = -2$, $z = 4$, $x = 3$, so $a + b + c = 5$.
4	**(e)**	$\Phi^{-1}\mathbf{f} = (2, -3e^{-t})$, $X_p = ((4t+3)e^t, (2t+3)e^t)$, $X_p(1) = (7e, 5e)$.
5	**(c)**	$\lambda^3 - 4\lambda^2 + 5\lambda - 2 = 0$, so $A^4 = 4A^3 - 5A^2 + 2A$ and $a + b + c = 1$.
6	**(b)**	$A^3 = 0$: $e^{At} = I + At + \frac{t^2}{2}A^2$, $f = 3t$, $h = 4t + \frac{9}{2}t^2$; $6 + 26 = 32$.
7	**(d)**	$W = -6$, $u_1' = \frac{3}{2}xe^{-6x}$, $u_2' = -\frac{3}{2}x$: $y_p = u_1e^{3x} + u_2e^{-3x}$.
8	**(e)**	Eigenpairs: $1 \to (1,1)$, $3 \to (4,3)$, so $P = \begin{bmatrix} 1 & 4 \\ 1 & 3 \end{bmatrix}$.
9	**(b)**	$x_1 = x, x_2 = x', x_3 = x''$: $x_1' = x_2$, $x_2' = x_3$, $t^3x_3' = -5x_1 - 3tx_2 + 2t^2x_3 + \ln t$.
10	**(b)**	Eigenvalues $3, -4, 0$; $\lambda = 3$: $(2,3,-2)$, $\lambda = -4$: $(-1,2,1)$; product -24.
11	**(b)**	$v_2 = (0,0,1)$, $v_1 = (1,1,0)$; solutions v_1, $v_2 + tv_1$, $v_3 + tv_2 + \frac{t^2}{2}v_1$ times e^{-t}.
12	**(c)**	$\lambda = 1 \pm 2i$: $X = e^t[(\cos 2t, \sin 2t) - 2(\sin 2t, -\cos 2t)]$, at $\frac{\pi}{4}$: $(-2,1)e^{\pi/4}$.
13	**(a)**	Free b, c, d: $(b + c + d, b, c, d)$ gives $(1,1,0,0), (1,0,1,0), (1,0,0,1)$.
14	**(d)**	$r^3 + 3r^2 - 4 = (r-1)(r+2)^2$: $y = c_1e^x + c_2e^{-2x} + c_3xe^{-2x}$.
15	**(d)**	Separable: $y^2 = x^2 + x - 1$; $y(-2) < 0$ gives $y(2) = -\sqrt{5}$.
16	**(e)**	$\lambda = 5$: multiplicity 3, 2 eigenvectors, so blocks of size 2 and 1.
17	**(c)**	$\rho = x^{-3}$: $(x^{-3}y)' = \frac{1}{x}$, so $y = x^3 \ln x + cx^3$.
18	**(b)**	$(A - 2I)\mathbf{v} = 0$: $z = 2y$, $x = -y$, so $\mathbf{v} = (-1,1,2)^T$.
19	**(c)**	$F = x^3 + 2xy^2 + g(y)$, $g' = 6y^2$: $x^3 + 2xy^2 + 2y^3 = c$.
20	**(d)**	$y = \cos 2x + \frac{3}{4}\sin 2x + \frac{x}{2}$, so $y(\pi) = 1 + \frac{\pi}{2}$.